# B28: SQL SuperHero Program : Session 25

1 message

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## LISTAGG

The Oracle LISTAGG() function is an aggregation function that transforms data from multiple rows into a single list of values separated by a specified delimiter.

```
LISTAGG (
    column_name [,
    delimiter]
) WITHIN GROUP(
    ORDER BY
        sort_expressions
);
```

- The column_name can be a column, constant, or expression that involves the columns.
- The delimiter is a string that separates the values in the result row. The delimiter can be NULL , a string literal, bind variable, or constant expression. If you omit the delimiter, the function uses a NULL value by default.
- The sort_expressions is a list of sort expressions to sort data in ascending (ASC) or descending (DESC) order. Note that you can use NULLS FIRST or NULLS LAST in the sort_expression to control the sort order of NULLs. By default, the LISTAGG() function uses ASCENDING and NULLS LAST options

-------------------------------------------------------

```
create table employ (id number, dept_id varchar2(20));

insert into employ values (10,'IT');
insert into employ values (20,'IT');
insert into employ values (30,'IT');
insert into employ values (40,'HR');
insert into employ values (50,'HR');
insert into employ values (60,'Admin');
insert into employ values (70,'Admin');
insert into employ values (80,'Admin');
```

Worksheet | Query Builder

```sql
select * from employ
```

Script Output | Query... 

SQL | All Rows Fetched: 8 in 0.003 seconds

| | ID | DEPT_ID |
|---|---|---|
| 1 | 10 | IT |
| 2 | 20 | IT |
| 3 | 40 | HR |
| 4 | 50 | HR |
| 5 | 60 | Admin |
| 6 | 70 | Admin |
| 7 | 80 | Admin |
| 8 | 30 | IT |

Expect to convert in single row

**Expected output:**

IT 10,20,30
HR 40,50
Admin 60,70,80

**select dept_id, listagg(id,',') within group (order by id) from employ group by dept_id;**

delimiter is , (comma)

Worksheet | Query Builder

```sql
select dept_id, listagg(id,',') within group (order by id) from employ group by dept_id;
```

this column, we want to comma separated

Script Output | Query Result

SQL | All Rows Fetched: 3 in 0.008 seconds

| | DEPT_ID | LISTAGG(ID,',')WITHINGROUP(ORDERBYID) |
|---|---|---|
| 1 | Admin | 60,70,80 |
| 2 | HR | 40,50 |
| 3 | IT | 10,20,30 |

**--> Whenever Convert Multiple rows into single row --> means Listagg**

```
select dept_id, listagg(id,'&') within group (order by id) from employ group by dept_id
```

'&' — seperated

| DEPT_ID | LISTAGG(ID,'&')WITHINGROUP(ORDERBYID) |
|---|---|
| Admin | 70&80 |
| HR | 40&50&60 |
| IT | 10&20&30 |

3 rows selected.

```
select dept_id, listagg(id,'#') within group (order by id) from employ group by dept_id
```

# seperated

| DEPT_ID | LISTAGG(ID,'#')WITHINGROUP(ORDERBYID) |
|---|---|
| Admin | 70#80 |
| HR | 40#50#60 |
| IT | 10#20#30 |

3 rows selected.

```
select dept_id, listagg(id,' ') within group (order by id) from employ group by dept_id
```

Space seperated

| DEPT_ID | LISTAGG(ID,'')WITHINGROUP(ORDERBYID) |
|---|---|
| Admin | 70 80 |
| HR | 40 50 60 |
| IT | 10 20 30 |

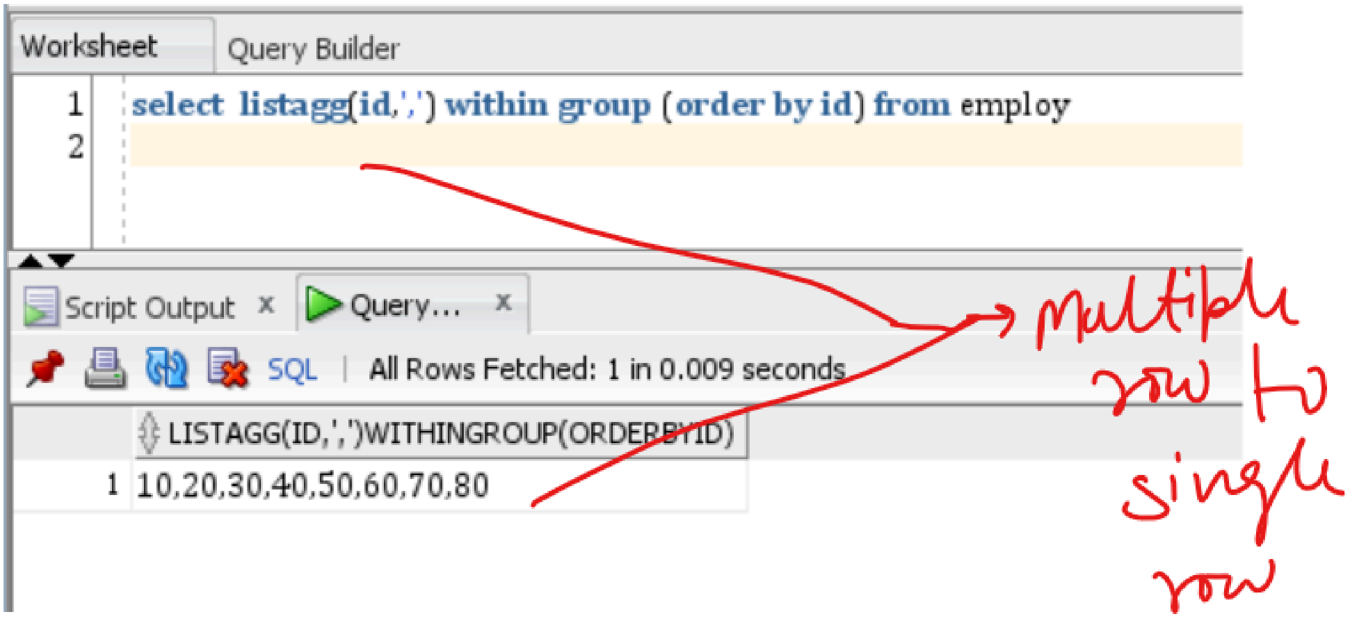
3 rows selected.

```
select dept_id, listagg(id,'XXXXXX') within group (order by id) from employ group by dept_id
```

seperated

| DEPT_ID | LISTAGG(ID,'XXXXXX')WITHINGROUP(ORDERBYID) |
|---|---|
| Admin | 70XXXXXX80 |
| HR | 40XXXXXX50XXXXXX60 |
| IT | 10XXXXXX20XXXXXX30 |

3 rows selected.




===================================================================================

```
SQL> select * from GfG;

SUBNO      SUBNAME
---------- -------------------------
D20        Algorithm
D30        DataStructure
D30        C
D20        C++
D30        Python
D30        DBMS
D10        LinkedList
D20        Matrix
D10        String
D30        Graph
D20        Tree

11 rows selected.
```

**Write an SQL query using LISTAGG function to output the subject names in a single field with the values comma delimited.**

SELECT LISTAGG(SubName, ' , ') WITHIN GROUP (ORDER BY SubName) AS SUBJECTS FROM   GfG ;

**Write an SQL query to group each subject and show each subject in its respective department separated by comma**

**Each means --> group by**

**SELECT SubNo, LISTAGG(SubName, ' , ') WITHIN GROUP (ORDER BY SubName) AS SUBJECTS FROM   GfG GROUP BY SubNo;**

```
SUBNO     SUBJECTS
------    ----------------------------------------
D10       LinkedList , String
D20        Algorithm , C++ , Matrix , Tree
D30        C , DBMS , DataStructure , Graph , Pyth
```

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-

**LISTAGG functions have UNIQUE features from oracle 19C onwards.**
For example, if we have duplicate data and we convert those records from multiple rows to si angle, LISTAGG is not able to make it a distinct collection till 18C.

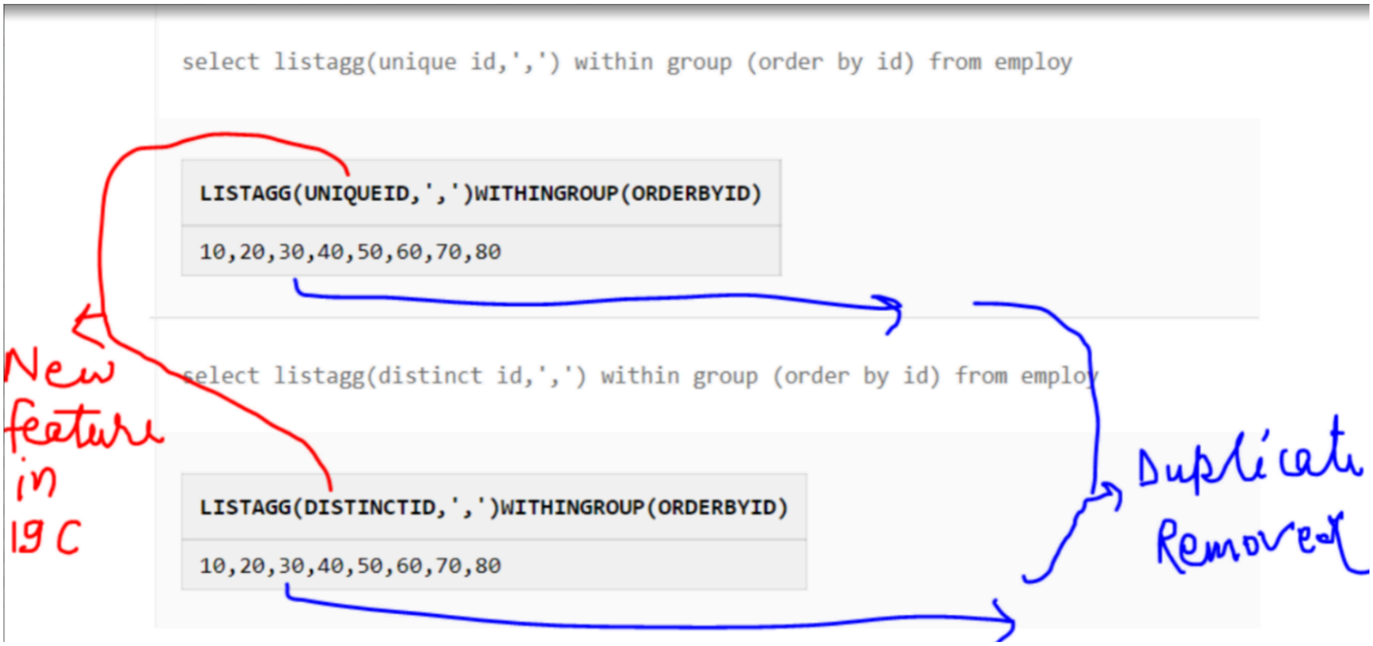
---

```
select listagg(id,',') within group (order by id) from employ
```

| LISTAGG(ID,',')WITHINGROUP(ORDERBYID) |
|---|
| 10,20,30,30,40,50,60,70,80 |

Duplicate Value

In 19C onwards, we can use unique. distinct keyword to pick unique records only.

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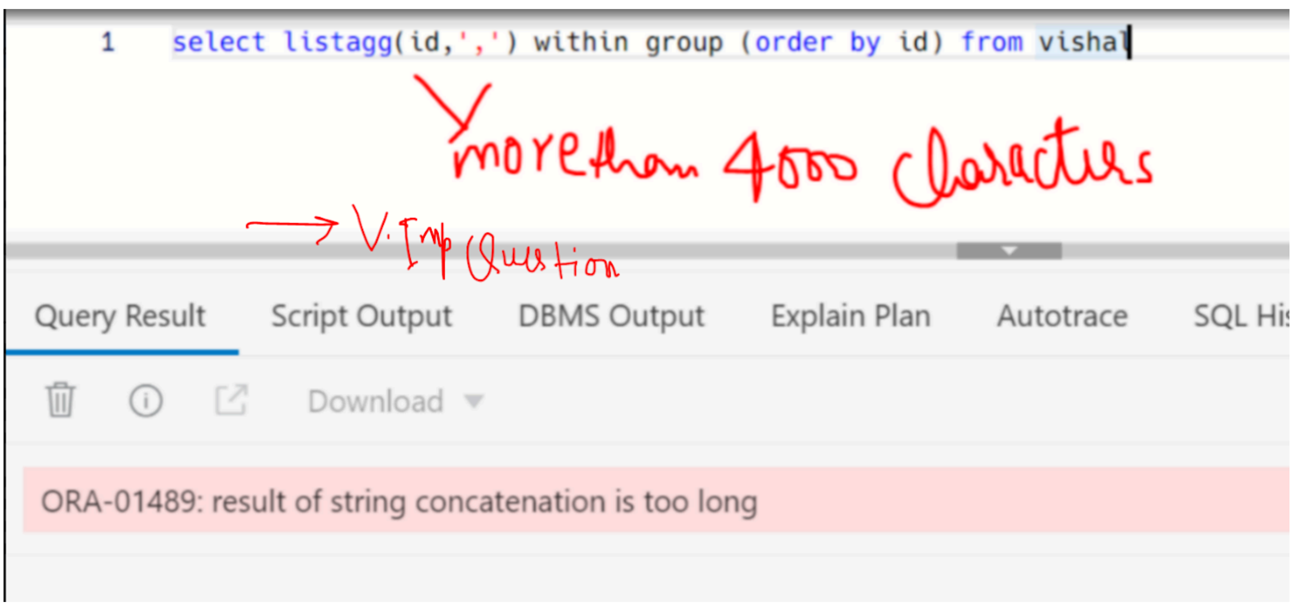
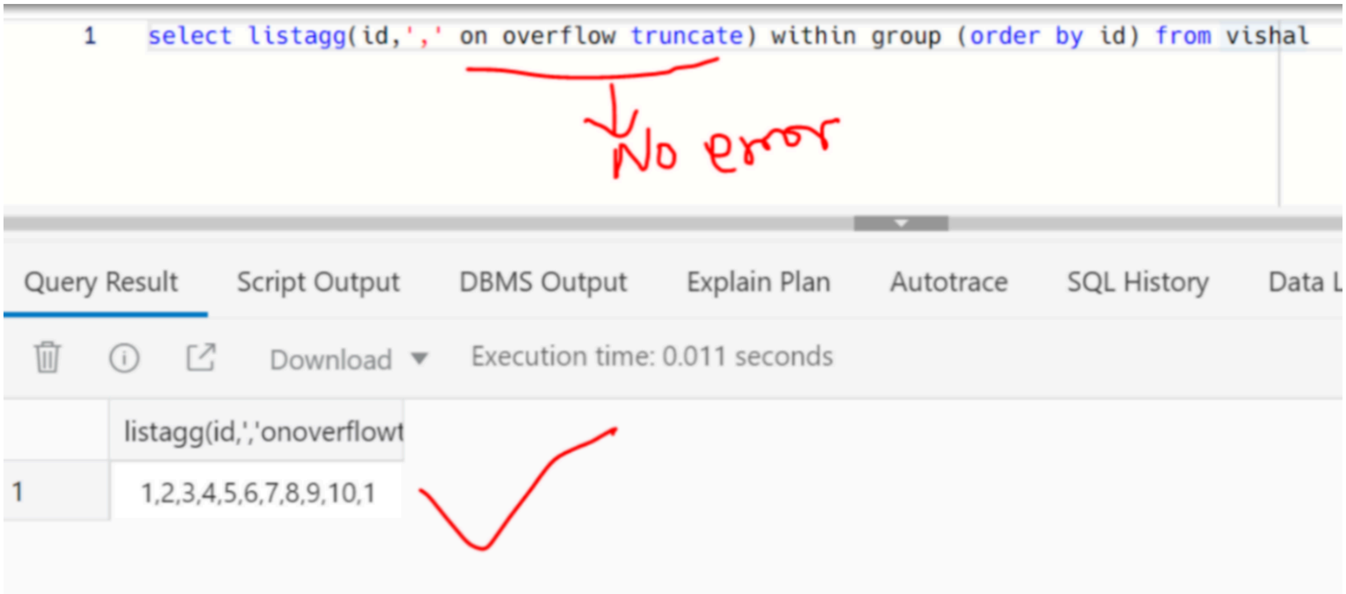
**We have a limit of 4000 bytes in listagg, if more than 4000 then it gives error:**

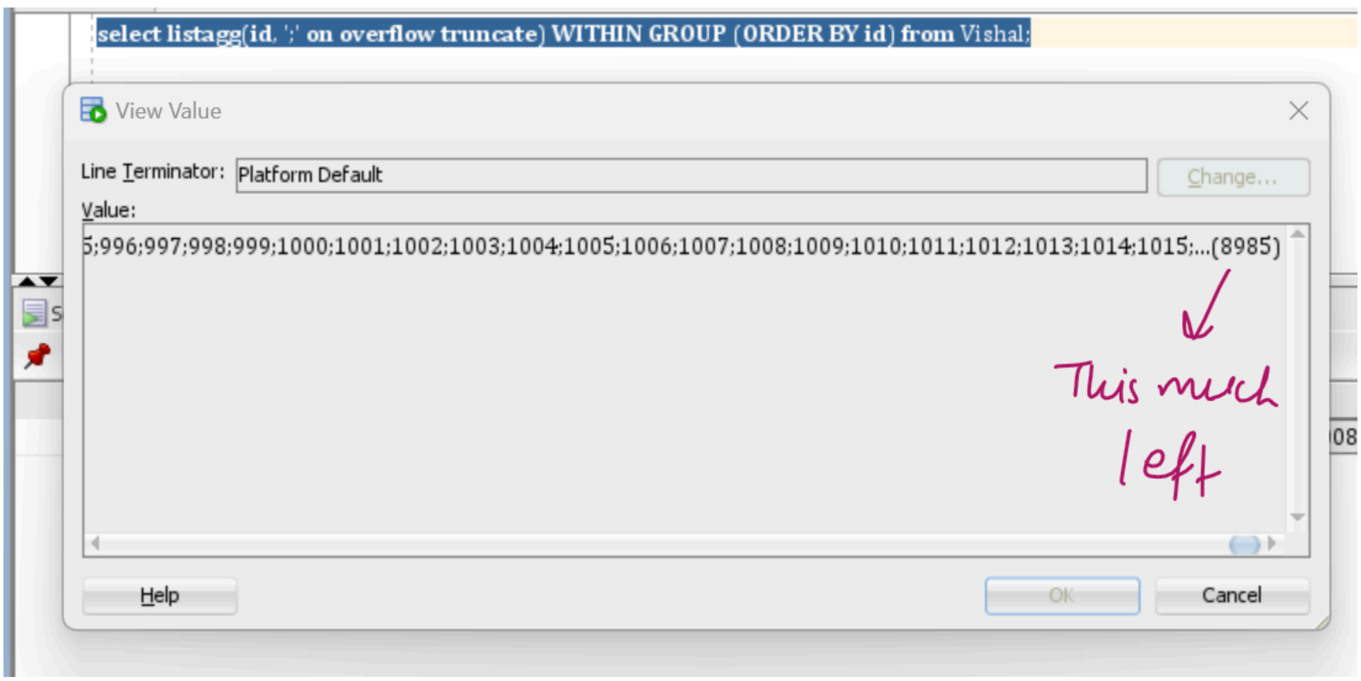
**Remember, Listagg function return is varchar. it returns a string.**



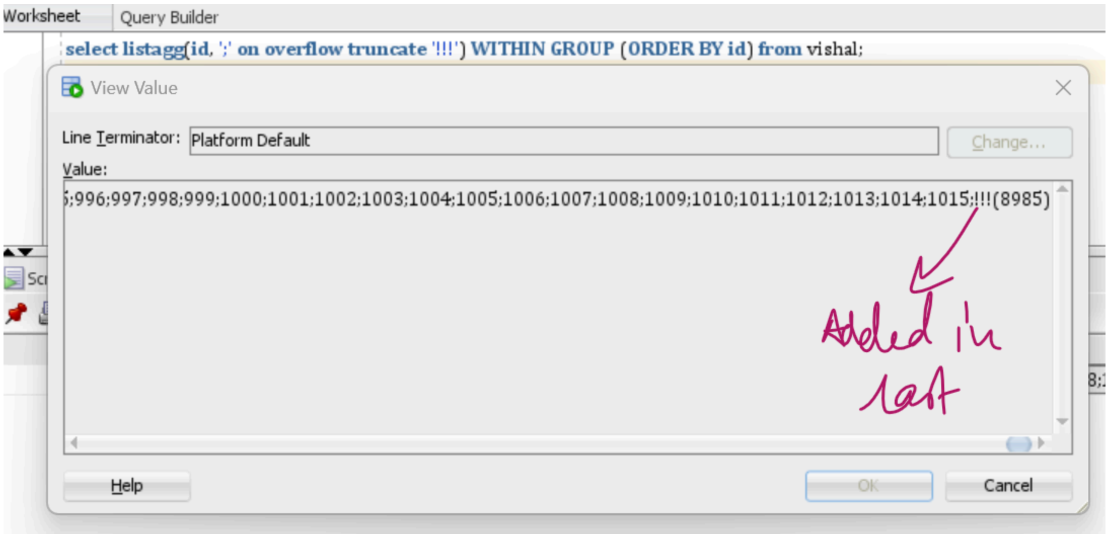

Solution for this is : "**on overflow truncate**" and it is only from oracle 12c onwards
In Oracle 12c R2, LISTAGG function has been enhanced to manage situations where the length of the concatenated string is too long. In Oracle Database 12c R2 provides us ON OVERFLOW TRUNCATE clause to handle ORA-01489 with LISTAGG.

**select listagg(id, ';' on overflow truncate) WITHIN GROUP (ORDER BY id) from Vishal;**
--> This will give all records till 4000 characters and no issue, no error, but stop when reaches to limit of 4000

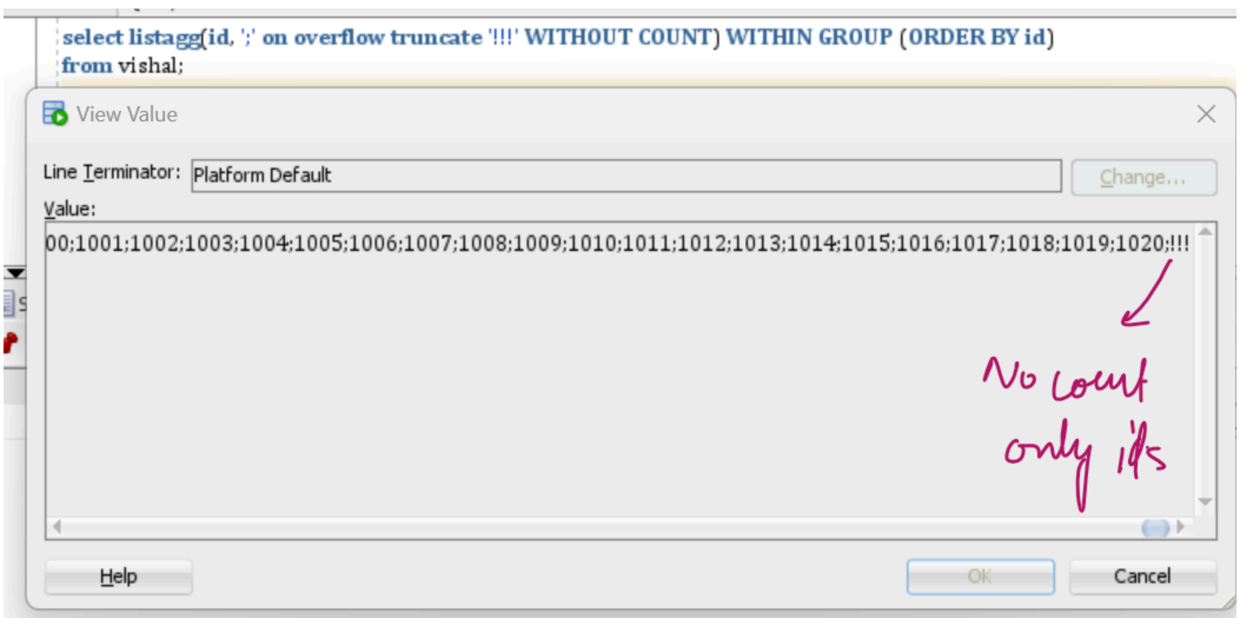



**select listagg(id, ';' on overflow truncate '!!!') WITHIN GROUP (ORDER BY id) from vishal;**
--> in last it will add !!!




**select listagg(id, ';' on overflow truncate '!!!' WITHOUT COUNT) WITHIN GROUP (ORDER BY id) from vishal;**
--> In last remove the total count of ids.

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**Can we write LISTAGG without order by**

**Yes**
select department_id, listagg(salary,',')  from hr.employees group by department_id

No order by → in oracle it support

```
select department_id,listagg(salary,',') datalist from hr.employees group by department_id
```

| DEPARTMENT_ID | DATALIST |
|---|---|
| 10 | 4400 |
| 20 | 13000,6000 |
| 30 | 11000,2500,2600,2800,2900,3100 |
| 40 | 6500 |
| 50 | 8000,2600,2600,3000,3100,2800,3200,3900,4000,2500,2900,3600,3800,3000,3400,4100,4200 |
| 60 | 9000,4200,4800,4800,6000 |
| 70 | 10000 |
| 80 | 14000,6200,8400,8600,8800,11000,6100,7300,7400,9600,10000,11500,6200,6400,6800,7200, |
| 90 | 24000,17000,17000 |
| 100 | 12008,6900,7800,7700,8200,9000 |
| 110 | 12008,8300 |
| - | 7000 |

**what if i do not want to pick data in any order**

**Order by null.**

```
select department_id,listagg(salary,',') within group (order by null) datalist from hr.employees group by department_id
```

means no order

| DEPARTMENT_ID | DATALIST |
|---|---|
| 10 | 4400 |
| 20 | 13000,6000 |
| 30 | 11000,2500,2600,2800,2900,3100 |
| 40 | 6500 |
| 50 | 8000,2600,2600,3000,3100,2800,3200,3900,4000,2500,2900,3600,3800,3000,3400,4100,4200,2800,2500,3100,320 |
| 60 | 9000,4200,4800,4800,6000 |
| 70 | 10000 |
| 80 | 14000,6200,8400,8600,8800,11000,6100,7300,7400,9600,10000,11500,6200,6400,6800,7200,9500,10500,7000,750 |
| 90 | 24000,17000,17000 |
| 100 | 12008,6900,7800,7700,8200,9000 |
| 110 | 12008,8300 |
| - | 7000 |

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what happens using the default delimiter

The result is the same, but the data is not separated by a comma. The default delimiter is nothing, so there is nothing in between each value.

```
1  select dept_id, listagg(id) within group (order by id) from employ
2  group by dept_id;
```

No delimiter given

| DEPT_ID | LISTAGG(ID)WITHINGROUP(ORDERBYID) |
|---|---|
| Admin | 607080 |
| HR | 4050 |
| IT | 102030 |

No delimiter may as it is data

Download CSV
3 rows selected.

==========================================================================================
=============================

```
create table entries (
name varchar2(20),
address varchar2(20),
email varchar2(20),
floor number,
resources varchar2(10));

insert into entries values('A','Bangalore','A@gmail.com',1,'CPU');
insert into entries values('A','Bangalore','A1@gmail.com',1,'CPU');
insert into entries values('A','Bangalore','A2@gmail.com',2,'DESKTOP');
insert  into entries values('B','Bangalore','B@gmail.com',2,'DESKTOP');
insert into entries values('B','Bangalore','B1@gmail.com',2,'DESKTOP');
insert into entries values('B','Bangalore','B2@gmail.com',1,'MONITOR');
```

input

| | NAME | ADDRESS | EMAIL | FLOOR | RESOURCES |
|---|---|---|---|---|---|
| 1 | A | Bangalore | A@gmail.com | 1 | CPU |
| 2 | A | Bangalore | A1@gmail.com | 1 | CPU |
| 3 | A | Bangalore | A2@gmail.com | 2 | DESKTOP |
| 4 | B | Bangalore | B@gmail.com | 2 | DESKTOP |
| 5 | B | Bangalore | B1@gmail.com | 2 | DESKTOP |
| 6 | B | Bangalore | B2@gmail.com | 1 | MONITOR |

Output

Query Result 2 | Query Result 3

All Rows Fetched: 2 in 0.046 seconds

| | NAME | TOTAL_VISIT | MOST_VISITED | RESOURCE_USED |
|---|---|---|---|---|
| 1 | A | 3 | 1 | CPU,DESKTOP |
| 2 | B | 3 | 2 | DESKTOP,MONITOR |

```
Understand the output. What the question wants. check the last column, means multiple row changed to
single row, means listagg.  we can write 1 line sql mfor this :
```

```
select name, listagg( resources,',') within group (order by resources) resource_used  from entries group by name
```

Query Result

All Rows Fetched: 2 in 0.011 seconds

| | NAME | RESOURCE_USED |
|---|---|---|
| 1 | A | CPU,CPU,DESKTOP |
| 2 | B | DESKTOP,DESKTOP,MONITOR |

But in output, duplicate not allowed, so lets use distinct or unique and remove the duplicates.

**select name, listagg(unique  resources,',') within group (order by resources) resource_used  from entries group by name**

```
select name, listagg(unique  resources,',') within group (order by resources) resource_used  from entries group by name
```

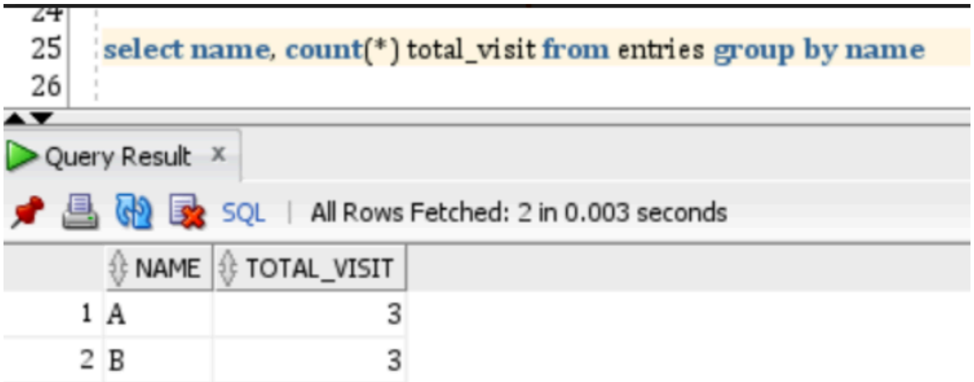
Query Result

All Rows Fetched: 2 in 0.003 seconds

| | NAME | RESOURCE_USED |
|---|---|---|
| 1 | A | CPU,DESKTOP |
| 2 | B | DESKTOP,MONITOR |

2 colums I got it

```
Now leave this and check the other column. How to get the total_visit column, actually it is just counter
the how many times A is there and how many times B is there.
```



```
select name, count(*) total_visit from entries group by name
```

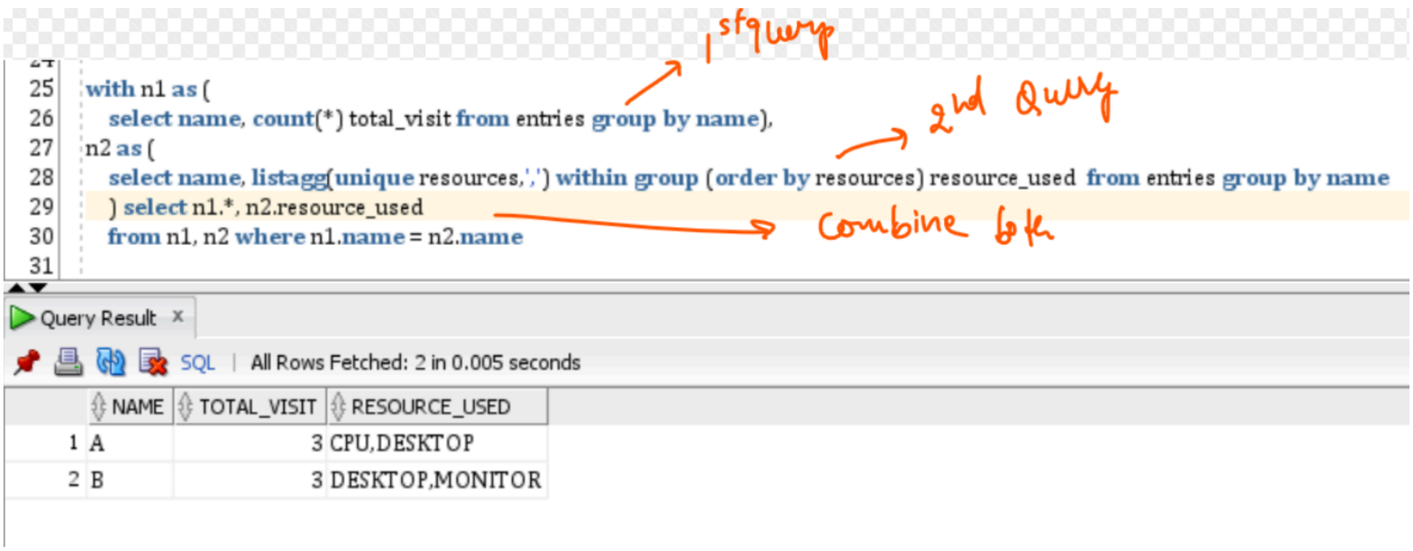
Query Result ×

SQL | All Rows Fetched: 2 in 0.003 seconds

| | NAME | TOTAL_VISIT |
|---|---|---|
| 1 | A | 3 |
| 2 | B | 3 |

lets combine both sql to get 3 column of output.

```
with n1 as (
   select name, count(*) total_visit from entries group by name),
n2 as (
   select name, listagg(unique resources,',') within group (order by resources) resource_used  from entries group by name
   ) select n1.*, n2.resource_used
   from n1, n2 where n1.name = n2.name
```



```
with n1 as (
   select name, count(*) total_visit from entries group by name),
n2 as (
   select name, listagg(unique resources,',') within group (order by resources) resource_used  from entries group by name
   ) select n1.*, n2.resource_used
   from n1, n2 where n1.name = n2.name
```



Query Result ×

SQL | All Rows Fetched: 2 in 0.005 seconds

| | NAME | TOTAL_VISIT | RESOURCE_USED |
|---|---|---|---|
| 1 | A | 3 | CPU,DESKTOP |
| 2 | B | 3 | DESKTOP,MONITOR |

Now just one column left, which is showing most visited. Like "A" visited floor 1 max times so output is 1 for A and "B" visited floor 2 max times, 2 for B.

```
select * from entries
```

Query Result

SQL | All Rows Fetched: 6 in 0.002 seconds

2 times

| | NAME | ADDRESS | EMAIL | FLOOR | RESOURCES |
|---|---|---|---|---|---|
| 1 | A | Bangalore | A@gmail.com | 1 | CPU |
| 2 | A | Bangalore | A1@gmail.com | 1 | CPU |
| 3 | A | Bangalore | A2@gmail.com | 2 | DESKTOP |
| 4 | B | Bangalore | B@gmail.com | 2 | DESKTOP |
| 5 | B | Bangalore | B1@gmail.com | 2 | DESKTOP |
| 6 | B | Bangalore | B2@gmail.com | 1 | MONITOR |

A

B

2 times

How to get it, just count and pick the maximum.

```
with n1 as (
   select name, count(*) total_visit from entries group by name),
n2 as (
   select name, listagg(unique resources,',') within group (order by resources) resource_used  from entries group by name
   ),
      N3 AS (
   SELECT NAME, FLOOR, COUNT(*) CN FROM entries GROUP BY NAME, FLOOR
   ) , N4 AS(SELECT n3.*, ROW_NUMBER() OVER(PARTITION BY NAME ORDER BY CN DESC) RN FROM N3)
   select N1.*, N2.resource_used, N4.FLOOR MOST_VISITED
   from N1,n2, N4
   WHERE N1.NAME = N2.NAME
   AND N1.NAME = N4.NAME
   AND N4.RN = 1
```

```
with n1 as (
   select name, count(*) total_visit from entries group by name),
n2 as (
   select name, listagg(unique resources,',') within group (order by resources) resource_used  from entries group by name
   ),
      N3 AS (
   SELECT NAME, FLOOR, COUNT(*) CN FROM entries GROUP BY NAME, FLOOR
   ) , N4 AS(SELECT n3.*, ROW_NUMBER() OVER(PARTITION BY NAME ORDER BY CN DESC) RN FROM N3)
   select N1.*, N2.resource_used, N4.FLOOR MOST_VISITED
   from N1,n2, N4
   WHERE N1.NAME = N2.NAME
   AND N1.NAME = N4.NAME
   AND N4.RN = 1
```

→ we did already

→ counted here

→ Arrange in order

→ Pick No. 1

Query Result

SQL | All Rows Fetched: 2 in 0.005 seconds

| | NAME | TOTAL_VISIT | RESOURCE_USED | MOST_VISITED |
|---|---|---|---|---|
| 1 | A | 3 | CPU,DESKTOP | 1 |
| 2 | B | 3 | DESKTOP,MONITOR | 2 |

**WE DISCUSSED MANY QUESTIONS IN OUR GROUP . YOU WILL GET THE SAME IN INTERVIEW. JUST PREPARE WELL**